

Multi-Agent Simulation and Crowdsourced Reviews for Urban Vitality Analysis: Insights from a Peri-Urban City

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1 INTRODUCTION

Urban vitality characterizes an urban space as lively and welcoming for people, while guaranteeing diverse functions such as growth, mobility, and diversity (Liu et al. 2022). Early approaches to characterizing urban vitality came from environmental psychology, using traditional methods like face-to-face interviews or online and in-person questionnaires (Lee et al. 2021). Recently, the growing availability of digital data flows on human behavior and the built environment such as mobile location data, satellite imagery, points of interest, public transport maps, and online reviews, combined with machine and deep learning methods, offer new perspectives for studying urban dynamics (Amer et al. 2025).

In this sense, travel flows generate patterns at both city and global scales that help us approach urban vitality as an indicator of the success and proper functioning of a public space. Beyond negative externalities, such as urban pollution or traffic congestion, analyzing travel across different modes (car, walking, biking, public transportation) enables us to determine the level of activity and attendance in public spaces. This approach uses multi-agent modeling via the MATSim simulator, which allows us to capture travel dynamics and the complex interactions between users and urban planning in fine detail. Applying this theoretical framework to the case of Loos-en-Gohelle, a former coal mining town, allows us to analyze urban vitality within a peri-urban community characterized by high spatial fragmentation, where agricultural land makes up 70% of the territory.

This research explores the estimation of urban vitality through public space attendance using MATSim simulations and its relationship with the density of digital traces left by people. Specifically, we correlate the flows from MATSim with the volume of geolocated online reviews from Google Maps to assess whether these MATSim simulation attendance is consistent with real-world attendance dynamics, as proxied by these digital traces.

2 METHODOLOGY

The methodology adopted in this study comprises three principal steps.

- **Generation of travel demand** : This stage aims to accurately represent the mobility features and behaviours of the study population at the departmental scale (Figure 1-a). It relies on the *eqasim* framework (Hörl et al. 2021), which enables the generation

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of synthetic agents, the assignment of daily activity chains, and the determination of the spatial location of these activities from local and open socio-demographic data. The department modelled is the Pas-de-Calais, a $120\text{km} \times 80\text{ km}$ area in the north of France of 1.4 Millions people.

- **Dynamic simulation via MATSim** : The generated mobility plans are fed into the multi-agent simulator *MATSim*. This framework models agent interactions on the transport network according to their individual preferences, evaluated by a calibrated scoring function. A "cutter" scenario, enabling the definition of the study area (here, Loos-en-Gohelle), is implemented in order to achieve a more detailed simulation (Figure 1-b). This step provides a realistic estimation of travel flows and public space attendance.
- **Analysis and correlation of volumes** : This step consists of aggregating the simulation results at the scale of urban sections. It enables the quantification of traffic volume through the spatial aggregation of passages observed on the links composing each section. These simulated flows are then linked to digital data extracted from social media (Google Maps reviews) in order to assess the correlation between the physical presence of people and their digital activity. The extraction of online reviews is performed using a Python script and the Selenium package, covering the period from March to April 2026.

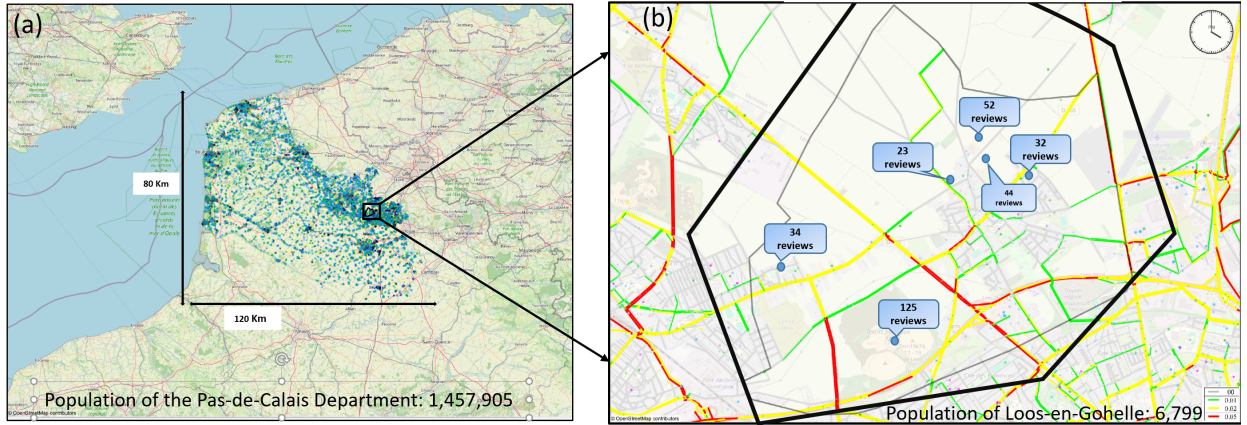


Figure 1: (a) Overview of agent-based simulation at the scale of the department. (b) Zoom-in on Loos-en-Gohelle surrounded by a spatial cutter, with traffic volumes and selected online review volumes.

In the absence of count data for simulation validation, this study uses geolocated reviews as an alternative. Following Li et al. 2025, the underlying assumption is that the volume of reviews constitutes a reliable proxy for actual attendance, with each review reflecting a physical visit to the point of interest. Furthermore, the volume of reviews has been demonstrated to be a valid proxy for mapping urban vitality (Noufe et al. 2026). Although review data is concentrated mainly on activity and service areas, it allows us to approach the dynamics of human presence in a context where validation data is missing.

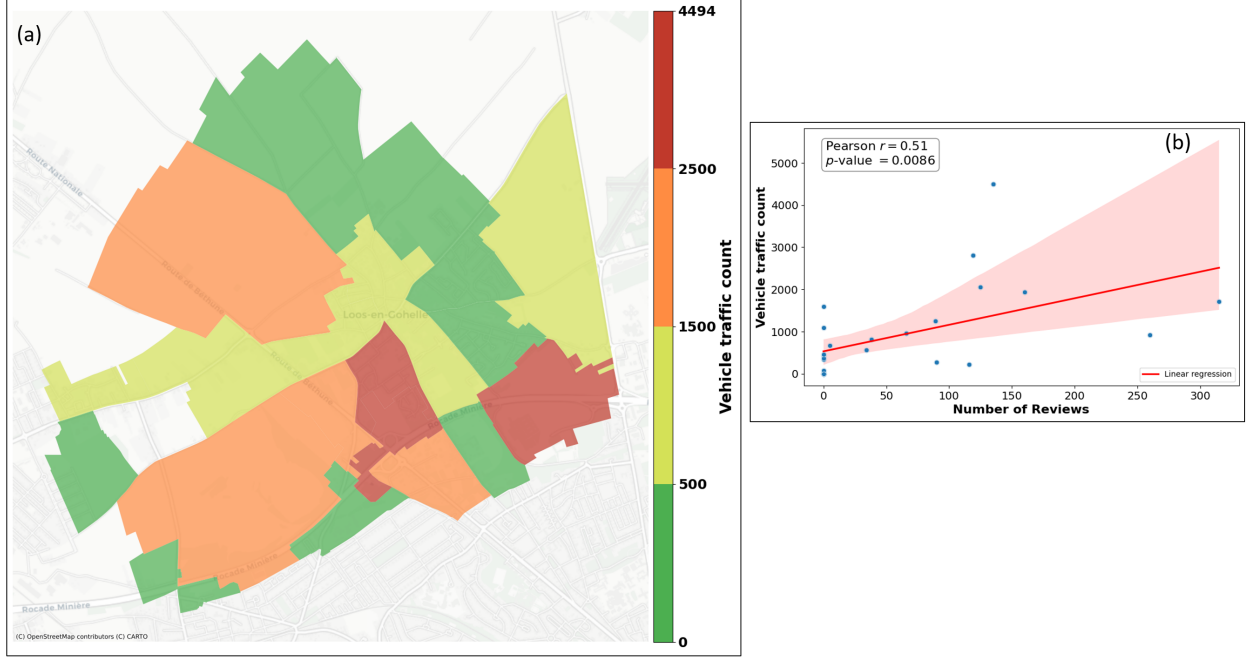


Figure 2: (a) Spatial distribution of vehicle traffic volume by cadastral section in Loos-en-Gohelle. (b) Pearson correlation between vehicle traffic volume and review volume in Loos-en-Gohelle.

3 RESULTS

We evaluate our approach on the commune of Loos-en-Gohelle, leveraging a travel demand model based on a synthetic population representing 10% of the real population, with a stabilised simulation after 32 iterations.

The distribution of vehicle travel flows at Loos-en-Gohelle (Figure 1-a) reveals an asymmetric structure in which maximum intensity ($> 2,500$ vehicles) is concentrated in the southeast, along major transportation corridors such as the Route de Béthune and the Rode Minière, rather than in the historic district. This heterogeneous configuration suggests that mobility is primarily driven by transit and commercial traffic, while the western and central-western residential areas experience heavy traffic volumes (1,500–2,500 vehicles), reflecting potentially significant commuting flows. By contrast, the northern and southwestern outskirts, characterised by low traffic (< 500 vehicles) and proximity to natural spaces such as the slag heap and agricultural areas, offer favourable conditions for soft mobility.

The intersection of vehicle volumes and the number of reviews by cadastral section (Figure 2-b) suggests a moderate Pearson correlation, statistically significant at the 5% threshold. This relationship indicates that an increase in automobile traffic tends to be associated with an increase in the number of reviews posted. Therefore user reviews appear to be correlated with the actual mobility dynamics simulated by MATSim.

Multi-agent simulation, and particularly the analysis of vehicle flows, constitutes a relevant approach to characterizing urban vitality, which can be validated against online review data. In our case, areas recording high travel volumes also exhibit high online review activity, and vice versa. This correspondence validates the use of travel demand models as proxies for analyzing urban vitality, although additional factors, such as urban morphology and functional diversity also contribute to shaping it.

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